

TOI-4145Ab

R-2768

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI4145_260730 · created 2026-09-03T22:55:44+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461251.979 +/- 0.07 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.051
Transit depth (Rp/Rs)^2	3.1 +/- 1.8 [%]
Orbital Inclination (inc)	88.65 +/- 2.54
Airmass coefficient 1 (a1)	0.107 +/- 0.046
Airmass coefficient 2 (a2)	1.34 +/- 1.41
Scatter in the residuals of the lightcurve fit is	42.56 %
Best Comparison Star	#2 - [221, 385]
Optimal Aperture	2.0
Optimal Annulus	6.07
Transit Duration (day)	0.037 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.051 vs 0.14195069566578392 (deviation 0.65σ)
Duration fit vs published	0.037 d vs 0.0875 d (deviation 1.5σ)
Mid-time O–C	130.8 min
Depth SNR	3.4
Residual scatter	42.56 %

Light curve

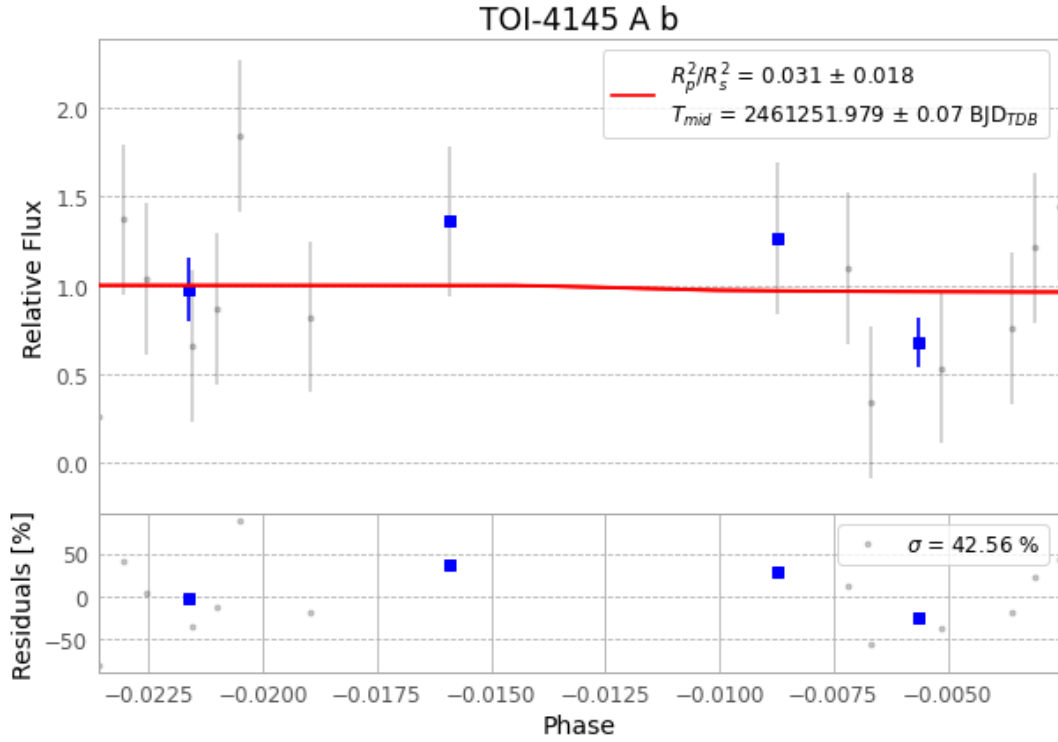


Figure 1 — FinalLightCurve_TOI-4145 A b_2026-07-30.png (SHA-256 f6461499d3be891a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 77], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4499d966145414a8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

42 FITS frames (28 MB), TOI4145260730091215.FITS → TOI4145260730111516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4145-260730.log (ab787a9cd584...), FinalLightCurve_TOI-4145 A b_2026-07-30.png (f6461499d3be...), FinalLightCurve_TOI-4145 A b_2026-07-30.csv (9b0c7e7ffe0f...)

Compute resources

Wall time 187.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 22:55Z.